MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 17: Difference-in-Differences (DiD) and Regression Discontinuity Designs (RDD)

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Part I: Introduction

Beyond Graphical Criteria

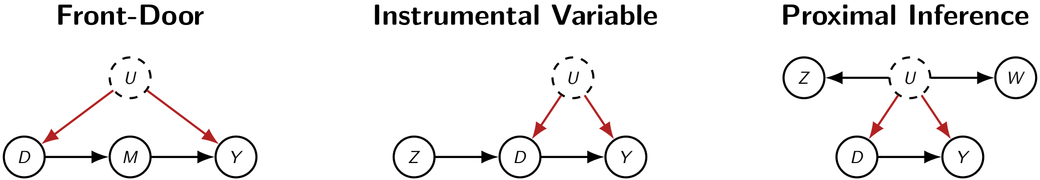
Goals of this Lecture

Today, we explore two powerful strategies for unobserved confounding that go beyond simple graphical criteria.

1. **Difference-in-Differences (DiD):** How can we use panel data (observations over time) to control for unobserved, time-invariant confounders?
2. **Regression Discontinuity (RDD):** How can we exploit sharp or fuzzy thresholds in treatment assignment to estimate local causal effects?
3. **Functional Form Assumptions:** Understand how these methods rely on assumptions about the *functional form* of the structural equations, not just the graph structure.

Recap: Three DAG-Based Strategies

We have seen three identification strategies that primarily required a particular DAG structure.



Identification from DAGs vs. Functional Form

Those methods worked because the graph satisfied key patterns:

- **Front-Door:** Required a *complete and unconfounded mediator* (M).
- **Instrumental Variables:** Required an *unconfounded instrument* (Z) with no direct effect on the outcome.
- **Proximal Inference:** Required two *conditionally independent proxies* (Z, W) for the confounder.

The partial linearity assumptions we added were mostly for **estimation**, not **identification**.

A New Kind of Assumption

Today's methods, DiD and RDD, cannot be deduced solely from the DAG. They rely on **functional form restrictions** in the structural equations, setting them qualitatively apart.

Part II: Difference-in-Differences

Comparing Changes Over Time

Motivating Example: A/B Testing with Confounding

Suppose you are Microsoft. For a subset of users who were most engaged with your product last quarter (based on a rich set of signals U), you deploy a new MS Office feature.

- We collect data on user activity, but the engagement signals U are not in the dataset.
- For each user (treated or not), we have their "active time spent on Office for each month of the year."
- The feature was released at the beginning of March.

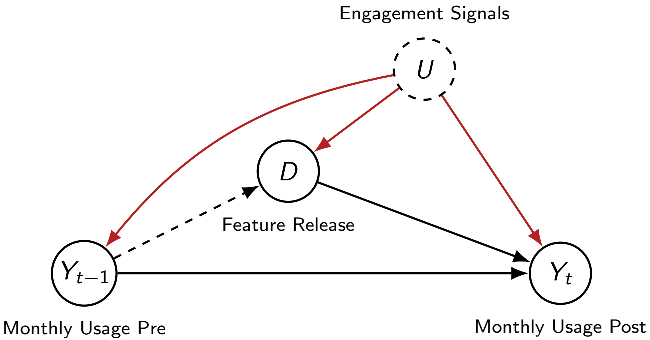
The Challenge

Can we estimate the effect of the release on the folks that were treated, even though treatment was assigned based on the unobserved confounder U?

The Causal Structure

Let's think about the causal relationships:

- High engagement (U) leads to higher baseline Office usage (Y_t).
- The new feature release (D) also affects Office usage (Y_t) after March.



The Core Idea: The Parallel Trends Assumption

We are willing to make a crucial assumption:

"Engagement signals that were used to determine who gets treated are indicative of who uses the product more (baseline level), but they are not indicative of whose usage will **grow** *more over time (trend), in the absense of treatment (baseline growth)."*

More formally, we assume that the average change in activity we would have observed from the treated group, *had they not been treated*, is the same as the average change in activity we observe from the control group.

The Parallel Trends Assumption (Formal)

Let $Y_t(0)$ be the potential outcome (usage) at time t without the feature.

Our assumption is that the expected *change* in the potential outcome under control is the same for both groups:

$$\mathbb{E}[Y_t(0) - Y_{t'}(0) \mid D = 1] = \mathbb{E}[Y_t(0) - Y_{t'}(0) \mid D = 0]$$

for a pre-treatment period t' and a post-treatment period t.

Key Insight

This is a form of **one-sided conditional ignorability**, but for the *change* in outcomes, not the levels.

From Parallel Trends to Identification

Poll Question

What quantity can we estimate when we have one-sided conditional ignorability?

From Parallel Trends to Identification

Poll Question

What quantity can we estimate when we have one-sided conditional ignorability?

Answer: The **Average Treatment Effect on the Treated (ATT)**.

Since we have one-sided ignorability for the *change* in outcomes, $\Delta Y_{t,t'} = Y_t - Y_{t'}$, we can identify the ATT *on the change*:

$$\text{ATT-on-Change} = \mathbb{E}[\Delta Y_{t,t'}(1) - \Delta Y_{t,t'}(0) \mid D = 1]$$

This is identified by:

$$\text{ATT-on-Change} = \mathbb{E}[\Delta Y_{t,t'} \mid D = 1] - \mathbb{E}[\Delta Y_{t,t'} \mid D = 0]$$

From ATT-on-Change to ATT-at-t

Is this what we wanted? We actually want the effect of the release in a specific post-treatment period t:

$$\text{ATT-at-t} = \mathbb{E}[Y_t(1) - Y_t(0) \mid D = 1]$$

Let's expand the quantity we just identified:

$$\begin{aligned}\text{ATT-on-Change} &= \mathbb{E}[Y_t(1) - Y_{t'}(1) - (Y_t(0) - Y_{t'}(0)) \mid D = 1] \\ &= \underbrace{\mathbb{E}[Y_t(1) - Y_t(0) \mid D = 1]}_{\text{ATT-at-t}} - \underbrace{\mathbb{E}[Y_{t'}(1) - Y_{t'}(0) \mid D = 1]}_{\text{Effect in pre-treatment period}}\end{aligned}$$

The No Anticipation Assumption

The second term, $\mathbb{E}[Y_{t'}(1) - Y_{t'}(0) \mid D = 1]$, is the treatment effect in a *pre-treatment* period t'.

We make a second key assumption:

The No Anticipation Assumption

The treatment, which happens at time $t_{release} > t'$, has no effect on outcomes in the pre-treatment period. Therefore:

$$Y_{t'}(1) = Y_{t'}(0)$$

This means the second term is zero, and we are left with:

$$\textbf{ATT-on-Change} = \textbf{ATT-at-t}$$

The Difference-in-Differences Estimator

We have just derived the DiD estimator!

The DiD Estimand for ATT

$$\text{ATT-at-t} = \underbrace{\mathbb{E}[Y_t - Y_{t'} \mid D = 1]}_{\text{Change in Treated}} - \underbrace{\mathbb{E}[Y_t - Y_{t'} \mid D = 0]}_{\text{Change in Control}}$$

We take the difference in outcomes before and after treatment for the treated group, and subtract the same difference for the control group.

Placebo Tests

How can we check the validity of our assumptions?

Placebo Test

Apply the same DiD methodology, but for two **pre-treatment** periods. The estimated "treatment effect" should be statistically zero.

If you find a non-zero effect, it's an indication that either:

- The parallel trends assumption is invalid.
- There are anticipation effects you didn't account for.

Empirical Example: GDPR and Firm Production

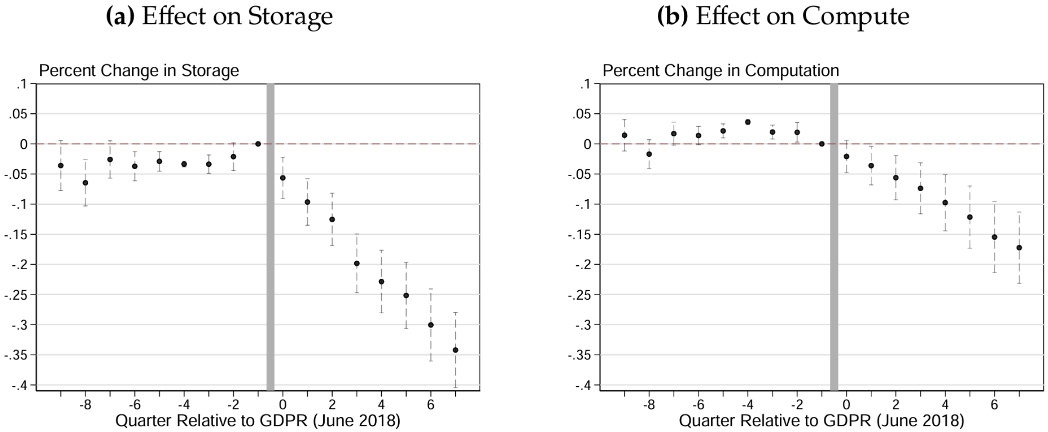
Demirer et al. (2024) wanted to understand the effects of GDPR on firm production, data usage, and IT investment.

- **Treatment:** GDPR, implemented in June 2018.
- **Treated Group:** EU firms (affected by GDPR).
- **Control Group:** US firms (not affected by GDPR).
- **Data:** Firm-level cloud usage data from a major global provider (July 2015 - March 2020).

Plausibility of Parallel Trends

It's plausible that US and EU firms have different baseline tech investment. However, if we believe their *growth* would have been the same absent GDPR, we can use DiD.

Demirer et al. (2024): Key Findings



Conditional Parallel Trends

What if the *composition* of firms in the EU and US is different based on observable characteristics X (e.g., by sector, size, etc.)?

Firms in different sectors can have different growth trajectories. This would violate parallel trends.

The Solution: Conditional Parallel Trends

We can assume that parallel trends holds *within strata* defined by X:

$$\mathbb{E}[\Delta Y_{t,t'}(0) \mid D = 1, X = x] = \mathbb{E}[\Delta Y_{t,t'}(0) \mid D = 0, X = x]$$

Estimation with Conditional Parallel Trends

This is one-sided conditional ignorability on the change in outcomes. We can use the g-formula for ATT:

$$\text{ATT} = \mathbb{E}[\Delta Y_{t,t'} \mid D = 1] - \mathbb{E}[g_0(X) \mid D = 1]$$

where $g_0(X) = \mathbb{E}[\Delta Y_{t,t'} \mid D = 0, X]$.

In practice:

1. Go to the control group (US firms) and learn a model $g_0(X)$ that predicts growth from firm characteristics.
2. Go to the treated group (EU firms) and for each firm with characteristics X, subtract the predicted growth $g_0(X)$ from its observed growth $\Delta Y_{t,t'}$.
3. Average this difference over all treated (EU firms).

Using Doubly Robust Estimation

Poll Question

If we want to use flexible ML methods to learn $g_0(X)$, what should we use instead of the simple g-formula?

Using Doubly Robust Estimation

Poll Question

If we want to use flexible ML methods to learn $g_0(X)$, what should we use instead of the simple g-formula?

Answer: The **Doubly Robust (DR) estimator for the ATT**.

- You have already implemented and used this in your homework!
- The only change is that you apply the DR estimator to the *change in outcomes* $\Delta Y_{t,t'}$ instead of the outcome levels Y_t.
- This allows you to estimate the ATT and its confidence interval for every post-treatment period.

Reminder: The Doubly Robust ATT Estimator

Recall from Lecture 4, the DR formula for ATT allows us to use ML for both the outcome and propensity models while getting valid CIs.

Population DR Formula for ATT

$$\text{ATT} = \frac{\mathbb{E}\Big[\, D\,(Y - g_0(X)) \;+\; (1 - D)\,\frac{p_0(X)}{1 - p_0(X)}\,(Y - g_0(X)) \,\Big]}{\mathbb{E}[D]}$$

- $g_0(X) = \mathbb{E}[Y \mid D = 0, X]$ is the outcome regression on controls.
- $p_0(X) = \Pr(D = 1 \mid X)$ is the propensity score.
- This estimator is **doubly robust**: it is consistent if either $\hat{g}_0$ or $\hat{p}_0$ is consistent (not both!) and is asymptotically normal under the *product rate condition*.

Pseudocode: Doubly Robust DiD Estimator

Simply define our outcome variable as the **change in outcomes**, $\Delta Y = Y_t - Y_{t'}$.

```python
# 1. Define the change in outcome
delta_Y = data['Y_post'] - data['Y_pre']

# 2. Use the standard DR ATT code from Lecture 4
#    with delta_Y as the outcome
cv = KFold(n_splits=5, shuffle=True, random_state=123)
g0hat, phat = np.zeros(n), np.zeros(n)
for train_idx, test_idx in cv.split(X):
    # Fit outcome model g(X) = E[delta_Y|D=0, X]
    untreated_train = train_idx[D[train_idx] == 0]
    g = clone(model_g).fit(X[untreated_train], delta_Y[untreated_train])
    g0hat[test_idx] = g.predict(X[test_idx])
    # Propensity p(x) = P(D=1|X=x)
    p = clone(model_p).fit(X[train_idx], D[train_idx])
    phat[test_idx] = p.predict_proba(X[test_idx])[:,1]

# 3. Compute the DR scores for ATT on the change
ipw_weight = phat / (1 - phat)
m = D * (delta_Y - g0hat) - (1 - D) * ipw_weight * (delta_Y - g0hat)

# 4. Compute DiD estimate and 95% CI
did_att = m.mean() / D.mean()
phi = (m - did_att * D) / D.mean()
se = phi.std() / np.sqrt(n)
ci95 = (did_att - 1.96 * se, did_att + 1.96 * se)
```

Placebo Tests

How can we check the validity of our assumptions?

Placebo Test

Apply the same DR-DiD methodology, but for two **pre-treatment** periods. The estimated "treatment effect" should be statistically zero.

If you find a non-zero effect, it's an indication that either:

- The **conditional parallel trends** assumption is invalid.
- There are anticipation effects you didn't account for.

Part III: Regression Discontinuity

Exploiting Thresholds

The RDD Idea: Treatment at a Threshold

Many treatments are administered based on a cutoff or threshold in a continuous variable (the *running variable*, X).

Example 1: Minimum Drinking Age

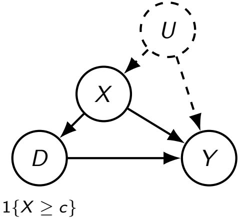
- **Treatment D:** Legal access to alcohol.
- **Running Variable X:** Age.
- **Cutoff c:** 21 years.
- Treatment is determined by a sharp rule: $D = \mathbf{1}\{\text{Age} \geq 21\}$.

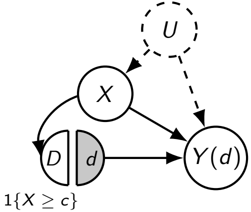
Example 2: Newborn ICU Care

- **Treatment D:** Receiving intensive ICU treatment.
- **Running Variable X:** Birth weight.
- **Cutoff c:** 1500 grams.
- Treatment is determined by a sharp rule: $D = \mathbf{1}\{\text{Birth Weight} \leq 1500g\}$.

The RDD Graph and the Identification Challenge

Let's draw the causal graph and SWIG for a Sharp RDD.





The Challenge: No Overlap!

Conditional exogeneity given X holds, but we have **zero overlap**. For any value of $X > c$, we only observe treated people and for any value of $X < c$ we only observe control, never both.

The Solution: The Smoothness Assumption

The key insight of RDD is to assume that the potential outcomes are **smooth** functions of the running variable.

"While individuals with $X = c + \epsilon$ and $X = c - \epsilon$ are different, as $\epsilon \to 0$, these individuals become arbitrarily similar in all ways, both observed and unobserved, **except for the treatment itself**.*"*

The Smoothness Assumption

The conditional expectations of the potential outcomes are continuous functions of x at the cutoff c.

$$\mathbb{E}[Y(d) \mid X = x] \text{ is continuous in } x \text{ at } c \text{ for } d \in \{0, 1\}$$

Identification in Sharp RDD: Step-by-Step

How does smoothness lead to identification?

1. For individuals just **below** the cutoff ($x < c$), they are all in the control group ($D = 0$). By conditional exogeneity and consistency:

$$\mathbb{E}[Y(0) \mid X = x] = \mathbb{E}[Y(0) \mid X = x, D = 0] = \mathbb{E}[Y \mid X = x, D = 0] = \mathbb{E}[Y \mid X = x]$$

2. By the smoothness assumption, as we approach the cutoff from below:

$$\lim_{x \uparrow c} \mathbb{E}[Y \mid X = x] = \mathbb{E}[Y(0) \mid X = c]$$

3. We do the same from the other side. For individuals just **above** the cutoff ($x > c$), they are all treated ($D = 1$):

$$\lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] = \mathbb{E}[Y(1) \mid X = c]$$

The Sharp RDD Estimand

We want to identify the Average Treatment Effect *at the cutoff* c:

$$\tau_{RDD} = \mathbb{E}[Y(1) - Y(0) \mid X = c]$$

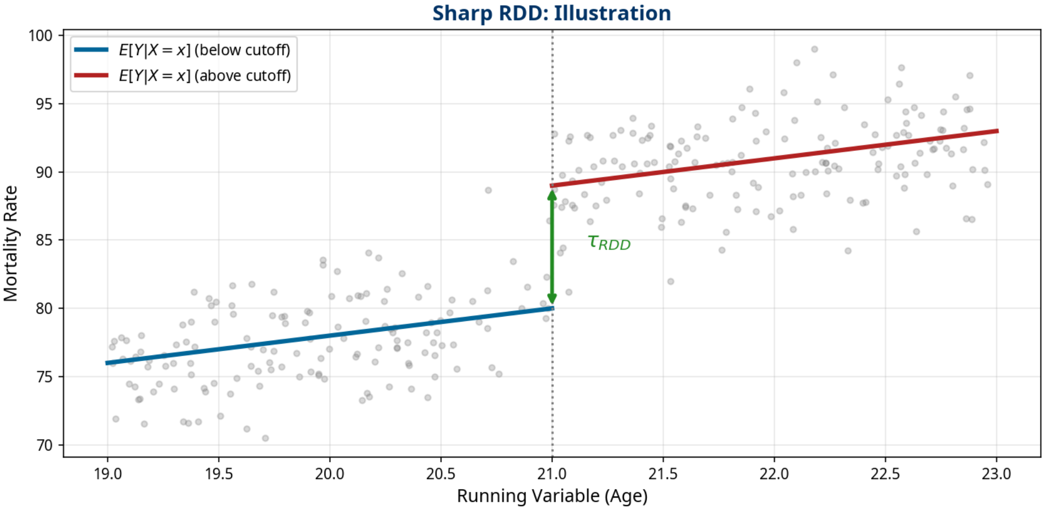
From the previous slide, we have:

$$\tau_{RDD} = \lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid X = x]$$

The Sharp RDD Estimand

The treatment effect at the cutoff is the size of the **jump** (discontinuity) in the observed outcome regression line at c.

The Sharp RDD Estimand Visually



Sharp RDD in Practice: Local Polynomial Regression

How do we operationalize this?

1. We learn a regression function $\hat{\mu}(x) = \mathbb{E}[Y|X = x]$ on either side of the cutoff.
2. This is typically done with **Local Linear (or Polynomial) Regression**, which fits a weighted regression in a small window (bandwidth h) around the cutoff.
3. The kernel function $K((X - c)/h)$ places more weight on samples near c.
4. We then take the difference in the predicted values from the two regressions right at the cutoff c.

Important note: If you have other covariates, you can add them as extra regressors in the local regression for **variance reduction**.

Bandwidth Selection and Robust Inference

A crucial choice is the **bandwidth** h. Too large, and you introduce bias by using dissimilar units. Too small, and your variance is too high.

- Standard methods choose h to minimize Mean Squared Error (MSE).
- This can introduce bias in the point estimate.
- State-of-the-art methods implemented in R and Python provide bias-correction and robust confidence intervals.

Key Reference

Calonico, Cattaneo, and Titiunik (2014). *Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs*. Econometrica.

Sharp RDD in Practice: rdrobust Package

For robust, data-driven bandwidth selection and bias-corrected CIs, we use the rdrobust package.

```python
from rdrobust import rdrobust, rdplot
import pandas as pd

# Load data (example from senate elections)
data = pd.read_csv("rdrobust_senate.csv")

# --- Sharp RDD ---
# y: outcome variable (vote share in next election)
# x: running variable (vote margin at current election)
# c: cutoff (0)
sharp_est = rdrobust(y=data["vote"], x=data["margin"], c=0)

# Print full summary table
print(sharp_est)

# --- Plotting ---
rdplot(y=data["vote"], x=data["margin"], c=0,
       title="Sharp RDD: Incumbency Advantage",
       x_label="Vote Margin", y_label="Next Vote Share")
```

Example: Minimum Drinking Age (Carpenter & Dobkin, 2009)

Question: Does granting legal access to alcohol at age 21 cause an increase in mortality?

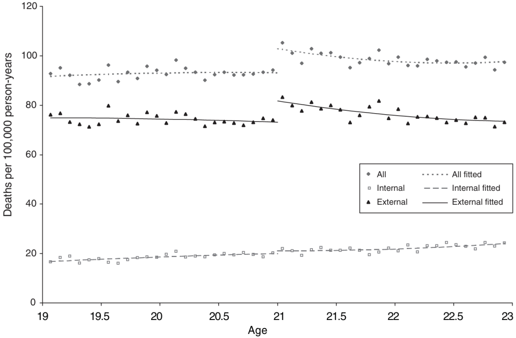
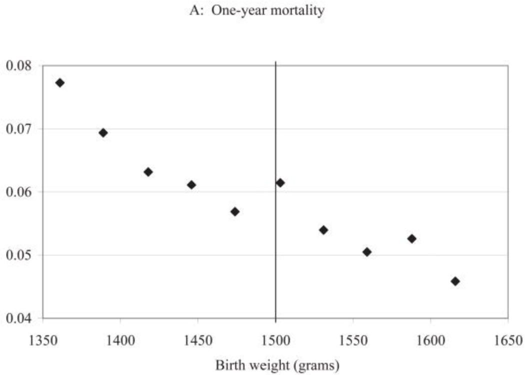


FIGURE 3. AGE PROFILE FOR DEATH RATES

- A **9%** increase in the overall mortality rate.
- A **15%** increase in deaths from motor vehicle accidents.

Example: At-Risk Newborns (Almond et al., 2010)

Question: What are the returns to medical care for newborns classified as `Very Low Birth Weight` (<1500g)?



- A 1 percentage point drop in one-year mortality (an **18% relative reduction**).

Part IV: Fuzzy RDD

When the Cutoff is Imperfect

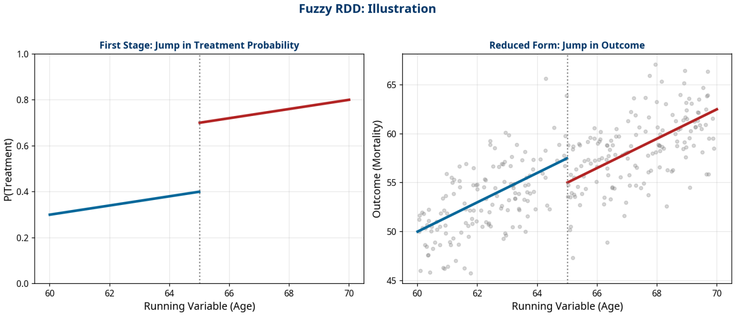
When the Cutoff Isn't Sharp

What if the threshold is not perfectly enforced?

- Some newborns with birth weight *above* 1500g still receive ICU treatment.
- Some newborns *below* the threshold do not.
- Eligibility for a program (e.g., financial aid, Medicare) doesn't guarantee take-up.

The cutoff creates a *discontinuity in the probability* of treatment, but this probability does not jump from 0 to 1.

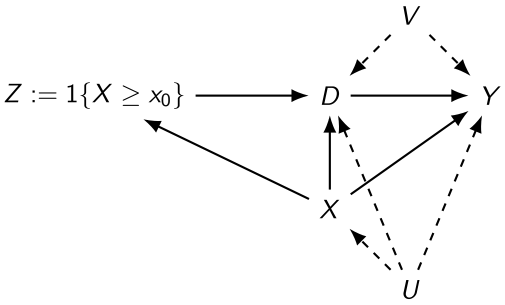
The Fuzzy RDD Visually



This is called a **Fuzzy Regression Discontinuity Design (Fuzzy RDD)**.

The Fuzzy RDD Causal Graph

Let's look at the causal graph. Let $Z = \mathbf{1}\{X \geq c\}$ be the indicator that the running variable exceeds the cutoff.



Poll Question

What familiar DAG does this resemble?

Fuzzy RDD as a Local IV

Answer: It's the **Instrumental Variable (IV)** graph!

In a Fuzzy RDD, we are essentially running an IV analysis in a narrow window around the cutoff.

- **Instrument:** $Z = \mathbf{1}\{X \geq c\}$. The discontinuity itself.
- **Treatment:** D, the treatment that is imperfectly assigned.
- **Outcome:** Y.

The RDD smoothness assumption ensures that the IV validity conditions hold *locally* at the cutoff:

- **Relevance:** The probability of treatment must be discontinuous at the cutoff.
- **Exclusion Restriction:** The cutoff Z only affects the outcome Y through its effect on the treatment D.

The Fuzzy RDD Estimand (Wald Estimator)

The effect is identified by the ratio of the jump in the outcome to the jump in the treatment probability (the first stage).

The Fuzzy RDD Estimand

$$\tau_{FuzzyRDD} = \frac{\lim_{x \downarrow c} \mathbb{E}[Y|X = x] - \lim_{x \uparrow c} \mathbb{E}[Y|X = x]}{\lim_{x \downarrow c} \mathbb{E}[D|X = x] - \lim_{x \uparrow c} \mathbb{E}[D|X = x]}$$

This is simply the **Wald estimator** for IV, applied to the discontinuities.

This identifies the **Local Average Treatment Effect (LATE)** for compliers at the cutoff.

Example: Financial Aid and Enrollment (Van der Klaauw, 2002)

Question: What is the causal effect of financial aid offers on college enrollment?

Design: Fuzzy RDD. Aid eligibility is determined by cutoffs in a composite academic index, but not all eligible students receive or accept aid.

Finding: Financial aid has a large and significant impact on enrollment.

- The elasticity of enrollment with respect to the grant amount is **0.86**.
- A $1,000 increase in the grant award increases the probability of enrollment by **8.6 percentage points** for students near the cutoff.

Fuzzy RDD in Practice: `rdrobust` Package

For a Fuzzy RDD, we simply add the `fuzzy` parameter, which specifies the treatment status variable.

```python
# --- Fuzzy RDD ---
# fuzzy: treatment variable (e.g., actual aid received)
# The package automatically calculates the Wald estimator.

# Example with hypothetical financial aid data
# Y: college_enrollment
# X: academic_index
# D: received_aid
c = 75 # Cutoff for aid eligibility

fuzzy_est = rdrobust(y=data["college_enrollment"],
                     x=data["academic_index"],
                     c=c,
                     fuzzy=data["received_aid"])

# Print full summary table
print(fuzzy_est)
```

Part V: Activity & Takeaways

Putting It All Together

Activity: Spot the Design

For each scenario, discuss with your neighbor:

1. Is this a DiD, Sharp RDD, or Fuzzy RDD problem?
2. What are the key variables (treatment, outcome, running variable/time)?
3. What is the key identifying assumption?

Activity: Scenario 1

Scenario 1: State-level minimum wage increase.

A state raises its minimum wage. To estimate the effect on employment, an economist collects monthly employment data from this state and a neighboring state that did not change its wage.

Activity: Scenario 1

Scenario 1: State-level minimum wage increase.

A state raises its minimum wage. To estimate the effect on employment, an economist collects monthly employment data from this state and a neighboring state that did not change its wage.

Answer

Design: Difference-in-Differences (DiD).

- **Treatment:** Minimum wage increase.
- **Outcome:** Employment level.
- **Key Assumption:** Parallel trends (employment growth would have been the same in both states absent the wage increase).

Activity: Scenario 2

Scenario 2: University scholarship program.

A university offers a merit scholarship to all applicants with an SAT score of 1400 or higher. We want to know the effect of the scholarship on enrollment.

Activity: Scenario 2

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A university offers a merit scholarship to all applicants with an SAT score of 1400 or higher. We want to know the effect of the scholarship on enrollment.

Answer

Design: Sharp Regression Discontinuity (Sharp RDD).

- **Treatment:** Receiving the scholarship.
- **Outcome:** Enrollment.
- **Running Variable:** SAT score.
- **Key Assumption:** Smoothness of potential outcomes at the 1400 SAT cutoff.

Activity: Scenario 3

Scenario 3: A get-out-the-vote campaign.

A campaign sends mailers to all registered voters in households with an average age of 50 or older. We want to know the effect of the mailer on voting, but we know some younger households might receive one by mistake and some older households might not.

Activity: Scenario 3

Scenario 3: A get-out-the-vote campaign.

A campaign sends mailers to all registered voters in households with an average age of 50 or older. We want to know the effect of the mailer on voting, but we know some younger households might receive one by mistake and some older households might not.

Answer

Design: Fuzzy Regression Discontinuity (Fuzzy RDD).

- **Treatment:** Receiving the mailer.
- **Outcome:** Voting.
- **Running Variable:** Average household age.
- **Key Assumption:** Smoothness of potential outcomes at the age 50 cutoff.

Takeaways

1. **Beyond DAGs:** DiD and RDD are powerful strategies for unobserved confounding that rely on **functional form assumptions** (parallel trends and smoothness) rather than purely on graph structure.
2. **Difference-in-Differences (DiD):** Compares the *change* in outcomes over time between treated and control groups. Identified by the **parallel trends assumption**, which can be made more plausible by conditioning on covariates (Conditional DiD) and checked with placebo tests.
3. **Regression Discontinuity (RDD):** Exploits a cutoff in a running variable to create a local randomized experiment. **Sharp RDD** assumes perfect compliance at the cutoff, while **Fuzzy RDD** handles imperfect compliance and is identified as a local IV.
4. **Local Effects:** RDD identifies the treatment effect *at the cutoff*, which may not generalize. DiD identifies the ATT for the treated population.

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